

XIAMEN UNIVERSITY MALAYSIA

厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Course Overview

By:

Dr. Ng Zi Neng

Email: zineng.ng@xmu.edu.my

Programme Educational Objective (PEO)

PEO1	**Professionalism** Graduates will establish themselves as practicing professionals in Electrical and Electronics Engineering or related fields.
PEO2	**Problem Analysis and Development of Solutions** Identify, formulate, conduct research literature and analyse complex engineering problems and design solutions that meets specified needs with appropriate considerations for public health and safely, cultural, societal and environmental considerations
PEO3	**Continuous Personal Development** Graduates will engage in lifelong pursuit of knowledge and interdisciplinary learning appropriate for industrial and academic careers.
PEO4	**Societal Engagement** Graduates will contribute to sustainable development and the well-being of society.

Course Learning Outcome (CLO) and Programme Learning Outcome (PLO)

CLO1	Describe basic physics, including mechanics and thermodynamics.
CLO2	Analyze and solve problems in engineering by physical principles.
CLO3	Apply the knowledge of physics to various engineering applications.

PLO1	Engineering Knowledge - Apply knowledge of mathematics, science, electrical & electronic engineering fundamentals and engineering specialisation principles to define and utilize engineering procedures, processes, systems and methodologies
PLO2	Problem Analysis - Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences
PLO3	Design/ Development of Solutions - Design solutions for complex engineering problems and systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

Timetable

- Lecture: Thursday, 4 – 6pm
- Tutorial: Friday, 4 – 5pm
- Week: 1-15 (20 Sept 2021 – 2 Jan 2022)
- Venue: MS Teams

  https://teams.microsoft.com/l/team/19%3a-0GUN9TEsa14jVfipur7Hb6SGLoBL-eLKypAMXqYO6w1%40thread.tacv2/conversations?groupId=f37ac5c5-d159-4c40-aa0f-62be755f58ab&tenantId=9e6a47d3-2795-4b97-9e52-78b1060629f1

Course Assessment

Assessment	Detail	(%)
Continuous Assessment	Assignments	10
	Quizzes	40
Final Assessment	Final exam (2 hours) Answer all questions	50
	Total	100

NO CHEATING POLICY

— Any academic misconduct is strictly prohibited. Strictly no copying allowed for assignments, quizzes and final exam.

—Discussion among students is encouraged for the assignments, but the final submission must be your own work.

Grading

Exam Questions, Marking Scheme & Invigilation	
Special Requirements (For EAC approved Engineering Programmes)	• Multiple Choice Questions (MCQ) are not encouraged in the final exams. • **Students shall pass the final examination to pass the course, the threshold marks for final examination is 40.**

Marks	Grade	Grade Points	Description
85 – 100	A	4.00	Distinction
80 – 84	A-	3.70	Distinction
75 – 79	B+	3.30	Good
70 – 74	B	3.00	Good
65 – 69	B-	2.70	Good
60 – 64	C+	2.30	Pass
55 – 59	C	2.00	Pass
50 - 54	C-	1.70	Conditional Pass
45 – 49	D+	1.30	Fail
40 - 44	D	1.00	Fail
≤ 39	F	0.00	Fail

Grading scales and grades applicable to ALL intakes effective from Academic session 2017/09

Course Plan

The two most basic and very important parts in Physics are Mechanics and Thermodynamics

1. Mechanics

- Week 1: Units, Physical Quantities, and Vectors
- Week 2: Motion Along A Straight Line (Motion in 1D)
- Week 3: Motion in 2D and 3D
- Week 4: Newton's Laws of Motion
- Week 5: Applying Newton's Laws of Motion
- Week 6: Revision (materials from Week 1 to Week 7)
- Week 7: Work, Kinetic Energy, Potential Energy, and Energy Conservation
- Week 8: Rotation of Rigid Bodies
- Week 9: Dynamics of Rotational Motion

Course Plan

The two most basic and very important parts in Physics are Mechanics and Thermodynamics

2. Thermodynamics

- Week 10: Temperature and Heat
- Week 11: Thermal Properties of Matter
- Week 12: The First Law of Thermodynamics
- Week 13: The Second Law of Thermodynamics
- Week 14: Revision for Mechanics
- Week 15: Revision for Thermodynamics

At the end of the course

You should be able:

- To apply basic laws of physics in mechanics and thermodynamics in various physical situations (problem-solving)
- To prepare for more advanced modules in Physics and other Sciences
- To think critically, ask the right questions, and articulate ideas clearly

Attendance

- Attendance is required by the Ministry of Higher Education Malaysia.
- In online mode, attendance will be recorded based on
  - your participation in live session or
  - your watch time on weekly pre-recorded video (on the same week).
- Sick leaves and other types of leaves should go through standard procedures, if you cannot make it to one of the above, else you will be marked as absent.
- For two consecutive absences, you will be reported to Academic Affairs.

How to learn Engineering Physics effectively

- Read the lecture slides and textbook prior to the lecture class/video
- Do not miss any class, pay attention and participate in the class
- Try the tutorials and examples from textbook frequently, practice makes perfect.
- Attempt the assignments by yourself, do not copy or let others to copy your work. Discussion is encouraged.

Textbook used in this course

University Physics with Modern Physics, 14th Edition by Hugh D Young, Roger Freedman.

- an excellent book and easy to understand
- many examples for course study

